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Areas of Expertise

Computer Vision, Deep Learning, Remote Sensing, Medical Imaging, Multimodal Integration

Contents

1 Education

2005–2010	Ph.D. in Computer Science Adviser: Robert Pless, Ph.D. Thesis: Calibrating and Using the Global Network of Outdoor Webcams	Washington University in St. Louis
1995–1999	B.S. in Computer Science (Minor in Mathematics) *Summa Cum Laude* with Honors	University of Missouri

2 Appointments and Affiliations

Active

Vice Provost for Artificial Intelligence — Washington University
May 2026– — *St. Louis, MO*

Professor (with Tenure) — Dept. of Computer Science & Engineering, Washington University
2022– — *St. Louis, MO*

Affiliated Faculty — Division of Computational & Data Sciences, Washington University
2022– — *St. Louis, MO*

Affiliated Faculty Imaging Science Program, Washington University
2022– *St. Louis, MO*

Affiliated Faculty Electrical and Systems Engineering, Washington University
2023– *St. Louis, MO*

Affiliated Faculty AI for Health Institute, Washington University
2023– *St. Louis, MO*

Associate Faculty Taylor Geospatial Institute
2022– *St. Louis, MO*

Faculty Scholar Center for the Environment, Washington University
2023– *St. Louis, MO*

Biodiversity Fellow Living Earth Collaborative
2022– *St. Louis, MO*

Technical Consultant / Owner / Founder Multidomain Vision Research, LLC
2019–present *St. Louis, MO*

Prior

Assistant Vice Provost for Digital Transformation Washington University
Jan 2025–May 2026 *St. Louis, MO*

Professor (with Tenure) Dept. of Computer Science, University of Kentucky
2021–2022 *Lexington, KY*

Director of Graduate Studies (Data Science) Dept. of Computer Science, University of Kentucky
2020–2022 *Lexington, KY*

Member Institute for Biomedical Informatics, University of Kentucky
2017–2022 *Lexington, KY*

Associate Professor (with Tenure) Dept. of Computer Science, University of Kentucky
2016–2021 *Lexington, KY*

Co-Department Chair (interim) Dept. of Computer Science, University of Kentucky
2019–2020 *Lexington, KY*

Affiliated Faculty Center for Visualization and Virtual Environments, University of Kentucky
2010–2019 *Lexington, KY*

Visiting Research Scientist (sabbatical) Orbital Insight, Inc.
2017–2018 *Mountain View, CA*

Assistant Professor Dept. of Computer Science, University of Kentucky
2010–2016 *Lexington, KY*

Computer Vision Research Intern ObjectVideo, Inc.
2008 *Reston, VA*

Graduate Research Assistant Dept. of Computer Science & Engineering, Washington University
2005–2010 *St. Louis, MO*

3 Awards

- Best Paper Award (out of 75 valid submissions) [EarthVision Workshop 2024 at IEEE/CVF Computer Vision and Pattern Recognition (CVPR)]
- Highlighted Reviewer Recognition (top 8%) [ICLR 2022]
- Outstanding Reviewer Recognition [BMVC 2021]
- Outstanding Reviewer Recognition (top 10%) [NeurIPS 2020]
- Outstanding Reviewer Recognition [ICCV 2019]
- University of Kentucky, College of Engineering Dean's Award for Excellence in Research [2018]
- Google Faculty Research Award [2018]
- Outstanding Reviewer Recognition [CVPR 2017]
- National Science Foundation CAREER Award [2016]
- Google Faculty Research Award [2016]
- Best Student Paper Award at Applied Imagery Pattern Recognition [2009]
- Ph.D. Forum Prize at the ACM/IEEE International Conference on Distributed Smart Cameras [2009]
- Best Talk Award for the Doctoral Student Seminar, Department of Computer Science, the Washington University in St. Louis, [Fall 2006]

4 Publications

Preprints

[1] S. Sastry, D. Cher, B. Wei, A. Dhakal, S. Khanal, D. Gupta, and N. Jacobs, "TerraDiT: Point-conditioned diffusion transformer for satellite image synthesis," in *TerraBytes — Towards global datasets and models for Earth observation (European Conference on Computer Vision (ECCV) Workshop)*, Sep. 2026. arXiv: 2603.02172 [cs.CV].

[2] Y. Cui, F. Qiao, and N. Jacobs, *StereoGenBench: A synthetic multi-camera benchmark for stereo generation under controlled baseline regimes*, May 2026. arXiv: 2605.23237 [cs.CV].

[3] C. Robinson, G. Muhawenayo, S. Khanal, Z. Fang, I. Corley, A. M. Tárano, L. Estes, J. Marcus, N. Jacobs, H. Kerner, I. Becker-Reshef, and J. M. L. Ferres, *The first global agricultural field boundary map at 10m resolution*, May 2026. arXiv: 2605.11055 [cs.CV].

[4] H. J. Koch, J. Zhu, S. Bhatt, N. Jacobs, and J. M. Dortch, "Assessing deep learning segmentation models for mapping landslides from lidar-based elevation data," Mar. 2026. DOI: 10.22541/essoar.177265426.62613754/v1 [Online]. Available: http://dx.doi.org/10.22541/essoar.177265426.62613754/v1

[5] Y. Luo, F. Qiao, Z. Xiong, Y. Li, and N. Jacobs, *GenOpticalFlow: A generative approach to unsupervised optical flow learning*, Mar. 2026. arXiv: 2603.22270 [cs.CV].

[6] Z. Xiong, Y. Song, L. He, W. Xiong, Y. Yuan, F. Qiao, and N. Jacobs, *PhysAlign: Physics-coherent image-to-video generation through feature and 3d representation alignment*, Mar. 2026. arXiv: 2603.13770 [cs.CV].

[7] F. Qiao, Z. An, Z. Xiong, S. Belongie, and N. Jacobs, *Track2View: 4d-consistent camera-controlled video generation via paired 3d point tracks*, 2026. arXiv: 2606.15534 [cs.CV].

[8] Z. Xiong, Y. Song, H. Kang, Q. Yan, L. Jiang, J. Yang, Z. Fu, S. Fotiadis, A. Wang, Z. Liu, B. Liu, Y. Yang, X. Lu, and N. Jacobs, *ActWorld: From explorable to interactive world model via action-aware memory*, 2026. arXiv: 2606.17730 [cs.CV].

[9] C. Mosig, T. Kattenborn, D. Montero Loaiza, J. Vanja-Jehle, J. Brandt, N. Jacobs, S. Khanal, E. Xing, M. Schwartz, H. C. Muller-Landau, M. Beloiu, A. Bozzini, Y. Cheng, K. Ganz, B. Grüning, H. Hartmann, J. Hempel, S. Horion, S. Junttila, K. Korznikov, G. Kraemer, M. Mönks, D. Nardi, P. Neumeier, J. Schmid, S. Soltani, M. Therese-Schmehl, J. Veitch-Michaelis, and M. Mahecha, "Sub-pixel mapping of disturbance and tree mortality dynamics from sentinel-2 time series around the globe," *Eartharxiv*, 2026. DOI: 10.31223/X5B18W [Online]. Available: https://eartharxiv.org/repository/view/11912/

[10] K. Klemmer, E. Rolf, M. Rußwurm, G. Camps-Valls, M. Czerkawski, S. Ermon, A. Francis, N. Jacobs, H. Kerner, L. Mackey, G. Mai, O. M. Aodha, M. Reichstein, C. Robinson, D. Rolnick, E. Shelhamer, V. Sitzmann, D. Tuia, and X. Zhu, "Earth embeddings: Toward artificial intelligence-centric representations of our planet," *IEEE Geoscience and Remote Sensing Magazine*, pp. 2–15, 2026. DOI: 10.1109/MGRS.2026.3710416

Refereed Conference Papers

[1] S. Khanal, Y. Cui, D. Cher, E. Xing, B. Wei, S. Sastry, and N. Jacobs, "Genesis: A generative engine for hierarchical satellite image synthesis," in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*, Nov. 2026.

[2] D. Cher, H. Iqbal, E. Xing, B. Wei, and N. Jacobs, "Tesselating the earth," in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: 2606.27514 [cs.CV].

[3] T. Gao, J. Lin, D. Beaulieu, and N. Jacobs, "Trajectory-aware cross-view geo-localization with sequential observations," in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: 2607.15491 [cs.CV].

[4] B. Wei, S. Sastry, D. Cher, E. Xing, and N. Jacobs, "TerraDiT-Ω: Unified spatial control for satellite image synthesis with any geospatial primitive," in *European Conference on Computer Vision (ECCV)*, Sep. 2026. eprint: 2606.31029 (cs.CV).

[5] Z. Xiong, X. Ye, B. Yaman, S. Cheng, Y. Lu, J. Luo, N. Jacobs, and L. Ren, "UniDrive-WM: Unified understanding, planning and generation world model for autonomous driving," in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: 2601.04453 [cs.CV].

[6] A. Dhakal, S. Khanal, S. Sastry, J. Arndt, P. A. Dias, D. Lunga, and N. Jacobs, "SimLBR: Learning to detect fake images by learning to detect real images," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026. arXiv: 2602.20412 [cs.CV].

[7] G. Muhawenayo, C. Robinson, S. Khanal, Z. Fang, I. Corley, A. Wollam, T. Gao, L. Strnad, R. Avery, L. Estes, A. M. Tárano, N. Jacobs, and H. Kerner, "PRUE: A practical recipe for field boundary segmentation at scale," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026.

[8] S. Sastry, S. Khanal, A. Dhakal, J. Lin, D. Cher, P. Jarosz, and N. Jacobs, "ProM3E: Probabilistic masked multimodal embedding model for ecology," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026. arXiv: 2511.02946 [cs.CV].

[9] A. Sarkar, S. Sastry, A. Pirinen, N. Jacobs, and Y. Vorobeychik, "DiffVAS: Diffusion-guided visual active search in partially observable environments," in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, (oral), May 2026. arXiv: 2605.15519 [cs.CV].

[10] D. Cher, B. Wei, S. Sastry, and N. Jacobs, "VectorSynth: Fine-grained satellite image synthesis with structured semantics," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Mar. 2026. arXiv: 2511.07744 [cs.CV].

[11] A. Wollam, K. Ashley, M. Shugaev, O. Arend, I. Y. Semenov, H. Dashtestani, S. Ravi, and N. Jacobs, "Towards unconstrained cross-view pose estimation," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Mar. 2026.

[12] A. Elallaf, N. Jacobs, X. Ye, M. Chen, and G. Liang, "Beta distribution learning for reliable roadway crash risk assessment," in *Association for the Advancement of Artificial Intelligence (AAAI)*, Jan. 2026. arXiv: 2511.04886 [cs.CV].

[13] E. Xing, A. Stylianou, R. Pless, and N. Jacobs, "QuARI: Query adaptive retrieval improvement," in *Neural Information Processing Systems (NeurIPS)*, vol. 2505.21647, Dec. 2025. arXiv: 2505.21647 [cs.CV].

[14] F. Qiao, Z. Xiong, E. Xing, and N. Jacobs, "Towards open-world generation of stereo images and unsupervised matching," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2025. arXiv: 2503.12720 [cs.CV].

[15] S. Sastry, A. Dhakal, E. Xing, S. Khanal, and N. Jacobs, "Global and local entailment learning for natural world imagery," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, vol. 2506.21476, Oct. 2025. arXiv: 2506.21476 [cs.CV].

[16] A. Dhakal, S. Sastry, S. Khanal, A. Ahmad, E. Xing, and N. Jacobs, "RANGE: Retrieval augmented neural fields for multi-resolution geo-embeddings," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025. arXiv: 2502.19781 [cs.CV].

[17] E. Xing, P. Kolouju, R. Pless, A. Stylianou, and N. Jacobs, "ConText-CIR: Learning from concepts in text for composed image retrieval," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025. arXiv: 2505.20764 [cs.CV].

[18] H. R. Kerner, S. Chaudhari, A. Ghosh, C. Robinson, A. Ahmad, E. Choi, N. Jacobs, C. Holmes, M. Mohr, R. Dodhia, J. M. L. Ferres, and J. Marcus, "Fields of The World: A machine learning benchmark dataset for global agricultural field boundary segmentation," in *Association for the Advancement of Artificial Intelligence (AAAI)*, vol. 2409.16252, Feb. 2025. arXiv: 2409.16252 [cs.CV].

[19] A. Sarkar, A. DiChristofano, S. Das, P. Fowler, N. Jacobs, and Y. Vorobeychik, "Active geospatial search for efficient tenant eviction outreach," in *Association for the Advancement of Artificial Intelligence (AAAI)*, Feb. 2025. arXiv: 2412.17854 [cs.LG].

[20] S. Sastry, S. Khanal, A. Dhakal, A. Ahmad, and N. Jacobs, "TaxaBind: A unified embedding space for ecological applications," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Feb. 2025. arXiv: 2411.00683 [cs.CV].

[21] M. Lanier, Y. Xu, N. Jacobs, C. Zhang, and Y. Vorobeychik, "Learning interpretable policies in hindsight-observable POMDPs through partially supervised reinforcement learning," in *IEEE International Conference on Machine Learning and Applications*, Dec. 2024. arXiv: 2402.09290 [cs.LG].

[22] A. Sarkar, S. Sastry, A. Pirinen, C. Zhang, N. Jacobs, and Y. Vorobeychik, "GOMAA-Geo: Goal modality agnostic active geo-localization," in *Neural Information Processing Systems (NeurIPS)*, Dec. 2024. arXiv: 2406.01917 [cs.CV].

[23] S. Khanal, E. Xing, S. Sastry, A. Dhakal, Z. Xiong, A. Ahmad, and N. Jacobs, "PSM: Learning probabilistic embeddings for multi-scale zero-shot soundscape mapping," in *ACM Multimedia*, Oct. 2024. DOI: 10.1145/3664647.3681620 arXiv: 2408.07050 [cs.CV].

[24] O. Skean, A. Dhakal, N. Jacobs, and L. G. S. Giraldo, "FroSSL: Frobenius norm minimization for self-supervised learning," in *European Conference on Computer Vision (ECCV)*, Oct. 2024. arXiv: 2310.02903 [cs.LG].

[25] A. Sarkar, A. DiChristofano, S. Das, P. Fowler, N. Jacobs, and Y. Vorobeychik, "Geospatial active search for preventing evictions," in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2024.

[26] C. Greenwell, M. Leotta, J. Crall, N. Jacobs, M. Purri, K. Dana, A. Hadzic, and S. Workman, "Watch: Wide-area terrestrial change hypercube," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.

[27] A. Sarkar, M. Lanier, S. Alfeld, J. Feng, R. Garnett, N. Jacobs, and Y. Vorobeychik, "A visual active search framework for geospatial exploration," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.

[28] S. Sastry, S. Khanal, A. Dhakal, D. Huang, and N. Jacobs, "BirdSat: Cross-view contrastive masked autoencoders for bird species classification and mapping," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.

[29] M. Shugaev, I. Semenov, K. Ashley, M. Klaczynski, N. Cuntoor, M. W. Lee, and N. Jacobs, "ArcGeo: Localizing limited field-of-view images using cross-view matching," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.

[30] A. Sarkar, N. Jacobs, and Y. Vorobeychik, "A partially-supervised reinforcement learning framework for visual active search," in *Neural Information Processing Systems (NeurIPS)*, Dec. 2023. arXiv: 2310.09689.

[31] S. Khanal, S. Sastry, A. Dhakal, and N. Jacobs, "Learning tri-modal embeddings for zero-shot soundscape mapping," in *British Machine Vision Conference (BMVC)*, Nov. 2023. arXiv: 2309.10667 [cs.CV].

[32] Z. Xiong, F. Qiao, Y. Zhang, and N. Jacobs, "StereoFlowGAN: Co-training for stereo and flow with unsupervised domain adaptation," in *British Machine Vision Conference (BMVC)*, Nov. 2023. arXiv: 2309.01842.

[33] X. Xing, C. Peng, Y. Zhang, A.-L. Lin, and N. Jacobs, "AssocFormer: Association transformer for multi-label classification," in *British Machine Vision Conference (BMVC)*, Nov. 2022.

[34] E. Xing, X. Xing, L. Liu, N. Jacobs, Y. Qu, and G. Liang, "Neural network decision-making criteria consistency analysis via inputs sensitivity," in *International Conference on Pattern Recognition (ICPR 2022)*, Aug. 2022. DOI: 10.1109/ICPR56361.2022.9956394

[35] S. Workman, M. U. Rafique, H. Blanton, and N. Jacobs, "Revisiting near/remote sensing with geospatial attention," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 25.33%, Jun. 2022. DOI: 10.1109/CVPR52688.2022.00182

[36] X. Xing, G. Liang, Y. Zhang, S. Khanal, A.-L. Lin, and N. Jacobs, "ADViT: Vision transformer on multimodality pet images for alzheimer disease diagnosis," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, Mar. 2022. DOI: 10.1109/ISBI52829.2022.9761584

[37] H. Blanton, S. Workman, and N. Jacobs, "A structure-aware method for direct pose estimation," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2022. DOI: 10.1109/WACV51458.2022.00028 arXiv: 2012.12360 [cs.CV].

[38] Y. Zhang, G. Liang, and N. Jacobs, "Dynamic feature alignment for semi-supervised domain adaptation," in *British Machine Vision Conference (BMVC)*, Nov. 2021. arXiv: 2110.09641.

[39] G. Liang, X. Xing, L. Liu, Y. Zhang, Q. Ying, A.-L. Lin, and N. Jacobs, "Alzheimer's disease classification using 2d convolutional neural networks," in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2021. DOI: 10.1109/EMBC46164.2021.9629587

[40] Q. Ying, X. Xing, L. Liu, A.-L. Lin, N. Jacobs, and G. Liang, "Multi-modal data analysis for Alzheimer's disease diagnosis: An ensemble model using imagery and genetic features," in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2021. DOI: 10.1109/EMBC46164.2021.9630174

[41] Y. Zhang, G. Liang, Y. Su, and N. Jacobs, "Multi-branch attention networks for classifying galaxy clusters," in *International Conference on Pattern Recognition (ICPR 2020)*, Acceptance rate: 28.47%, Jan. 2021. DOI: 10.1109/ICPR48806.2021.9412498

[42] G. Liang, Y. Zhang, X. Wang, and N. Jacobs, "Improved trainable calibration method for neural networks," in *British Machine Vision Conference (BMVC)*, Sep. 2020.

[43] M. U. Rafique, H. Blanton, N. Snavely, and N. Jacobs, "Generative Appearance Flow: A hybrid approach for outdoor view synthesis," in *British Machine Vision Conference (BMVC)*, Sep. 2020.

[44] G. Liang, X. Wang, Y. Zhang, and N. Jacobs, "Weakly-supervised self-training for breast cancer localization," in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, (oral), Jul. 2020. DOI: 10.1109/EMBC44109.2020.9176617

[45] T. Salem, S. Workman, and N. Jacobs, "Learning a dynamic map of visual appearance," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 25%, Jun. 2020. DOI: 10.1109/CVPR42600.2020.01245

[46] S. Workman and N. Jacobs, "Dynamic traffic modeling from overhead imagery," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 5.7% (oral), Jun. 2020. DOI: 10.1109/CVPR42600.2020.01233

[47] G. Liang, X. Wang, Y. Zhang, X. Xing, H. Blanton, T. Salem, and N. Jacobs, "Joint 2d-3d breast cancer classification," in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 18% (oral), Nov. 2019. DOI: 10.1109/BIBM47256.2019.8983048 arXiv: 2002.12392.

[48] Y. Zhang, X. Wang, H. Blanton, G. Liang, X. Xing, and N. Jacobs, "2d convolutional neural networks for 3d digital breast tomosynthesis classification," in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 18% (oral), Nov. 2019. DOI: 10.1109/BIBM47256.2019.8983097 arXiv: 2002.12314.

[49] G. Liang, S. Fouladvand, J. Zhang, M. A. Brooks, N. Jacobs, and J. Chen, "GANai: Standardizing CT images using generative adversarial network with alternative improvement," in *IEEE International Conference on Healthcare Informatics (ICHI)*, Jun. 2019. DOI: 10.1109/ICHI.2019.8904763

[50] Z. Bessinger and N. Jacobs, "A generative model of worldwide facial appearance," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, (oral), Jan. 2019. DOI: 10.1109/WACV.2019.00172

[51] N. Jacobs, A. Kraft, M. U. Rafique, and R. D. Sharma, "A weakly supervised approach for estimating spatial density functions from high-resolution satellite imagery," in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*, Acceptance rate: 22.5% (oral), Nov. 2018. DOI: 10.1145/3274895.3274934 arXiv: 1810.09528.

[52] R. P. Mihail and N. Jacobs, "Automatic hand skeletal shape estimation from radiographs," in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 19.6%, Nov. 2018. DOI: 10.1109/BIBM.2018.8621196

[53] S. Schulter, M. Zhai, N. Jacobs, and M. Chandraker, "Learning to look around objects for top-view representations of outdoor scenes," in *European Conference on Computer Vision (ECCV)*, Acceptance rate: 31.8%, Sep. 2018. DOI: 10.1007/978-3-030-01267-0_48 arXiv: 1803.10870.

[54] M. Zhai, T. Salem, C. Greenwell, S. Workman, R. Pless, and N. Jacobs, "Learning geo-temporal image features," in *British Machine Vision Conference (BMVC)*, Acceptance rate: 29.5%, Sep. 2018.

[55] C. Greenwell, S. Workman, and N. Jacobs, "What goes where: Predicting object distributions from above," in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Jul. 2018. DOI: 10.1109/IGARSS.2018.8519251 arXiv: 1808.00995.

[56] T. Salem, M. Zhai, S. Workman, and N. Jacobs, "A multimodal approach to mapping soundscapes," in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Jul. 2018. DOI: 10.1109/IGARSS.2018.8517977

[57] D. Jones, J. Bopaiah, F. Alghamedy, N. Jacobs, H. Weiss, W. A. D. Jong, and S. Ellingson, "Polypharmacology within the full kinome: A machine learning approach," in *AMIA Informatics Summit*, Mar. 2018.

[58] W. Song, S. Workman, A. Hadzic, R. Souleyrette, E. Green, M. Chen, X. Zhang, and N. Jacobs, "FARSA: Fully automated roadway safety assessment," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Mar. 2018. DOI: 10.1109/WACV.2018.00063 arXiv: 1901.06013.

[59] X. Zhang, Y. Zhang, E. Han, N. Jacobs, Q. Han, X. Wang, and J. Liu, "Whole mammogram image classification with convolutional neural networks," in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 19%, Nov. 2017. DOI: 10.1109/BIBM.2017.8217738

[60] N. Vo, N. Jacobs, and J. Hays, "Revisiting IM2GPS in the deep learning era," in *IEEE International Conference on Computer Vision (ICCV)*, Acceptance rate: 28.9%, Oct. 2017. DOI: 10.1109/ICCV.2017.286

[61] S. Workman, R. Souvenir, and N. Jacobs, "Understanding and mapping natural beauty," in *IEEE International Conference on Computer Vision (ICCV)*, Acceptance rate: 28.9%, Oct. 2017. DOI: 10.1109/ICCV.2017.596 arXiv: 1612.03142.

[62] S. Workman, M. Zhai, D. Crandall, and N. Jacobs, "A unified model for near and remote sensing," in *IEEE International Conference on Computer Vision (ICCV)*, Acceptance rate: 28.9%, Oct. 2017. DOI: 10.1109/ICCV.2017.293 arXiv: 1708.03035.

[63] M. Zhai, Z. Bessinger, S. Workman, and N. Jacobs, "Predicting ground-level scene layout from aerial imagery," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 29.2%, Jun. 2017. DOI: 10.1109/CVPR.2017.440

[64] Z. Bessinger, C. Stauffer, and N. Jacobs, "Who goes there? Approaches to mapping facial appearance diversity," in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*, Nov. 2016. DOI: 10.1145/2996913.2996997

[65] Z. Bessinger and N. Jacobs, "Quantifying curb appeal," in *IEEE International Conference on Image Processing (ICIP)*, Acceptance rate: 45%, Sep. 2016. DOI: 10.1109/ICIP.2016.7533189

[66] S. Workman, M. Zhai, and N. Jacobs, "Horizon lines in the wild," in *British Machine Vision Conference (BMVC)*, Acceptance rate: 39.4%, Sep. 2016. arXiv: 1604.02129.

[67] M. Zhai, S. Workman, and N. Jacobs, "Camera geo-calibration using an MCMC approach," in *IEEE International Conference on Image Processing (ICIP)*, Acceptance rate: 45%, Sep. 2016. DOI: 10.1109/ICIP.2016.7532905

[68] M. Zhai, S. Workman, and N. Jacobs, "Detecting vanishing points using global image context in a non-Manhattan world," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 29.9%, Jun. 2016. DOI: 10.1109/CVPR.2016.610 arXiv: 1608.05684.

[69] R. Baltenberger, M. Zhai, C. Greenwell, S. Workman, and N. Jacobs, "A fast method for estimating transient scene properties," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Acceptance rate: 42.3%, Mar. 2016, pp. 1–8. DOI: 10.1109/WACV.2016.7477713

[70] R. P. Mihail, S. Workman, Z. Bessinger, and N. Jacobs, "Sky segmentation in the wild: An empirical study," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Acceptance rate: 42.3%, Mar. 2016, pp. 1–6. DOI: 10.1109/WACV.2016.7477637

[71] T. Salem, S. Workman, M. Zhai, and N. Jacobs, "Analyzing human appearance as a cue for dating images," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Acceptance rate: 42.3%, Mar. 2016, pp. 1–8. DOI: 10.1109/WACV.2016.7477678

[72] C. Murdock, N. Jacobs, and R. Pless, "Building dynamic cloud maps from the ground up," in *IEEE International Conference on Computer Vision (ICCV)*, Acceptance rate: 30.3%, Oct. 2015, pp. 1–9. DOI: 10.1109/ICCV.2015.85

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Book Chapters

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Abstracts

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[2] D. Cher, P. Kolouju, E. Xing, R. Pless, A. Stylianou, and N. Jacobs, “Efficient hierarchical search for scalable retrieval in global remote sensing imagery,” in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, (oral), Jul. 2026.

[3] C. Mosig, T. Kattenborn, J. Vanja-Jehle, D. Montero, J. Brandt, A. Bozzini, Y. Cheng, A. Esquivel Muelbert, K. Ganz, E. Gora, B. A. Grüning, N. Jacobs, H. Hartmann, J. Hempel, S. Horion, S. Junttila, S. Khanal, K. Korznikov, G. Kraemer, I. McGregor, M. Mönks, H. Muller-Landau, D. Nardi, P. Neumeier, J. Schmid, M. Schwartz, M. Beloui Schwenke, S. Soltani, M. Therese-Schmehl, J. Veitch-Michaelis, E. Xing, and M. D. Mahecha, “Deadtrees.earth maps: Tree mortality and disturbance mapping from sentinel-2 timeseries across the globe,” in *EGU General Assembly 2026*, Vienna, Austria: Copernicus Meetings, May 2026. DOI: 10.5194/egusphere-egu26-18072 [Online]. Available: https://doi.org/10.5194/egusphere-egu26-18072

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[5] R. O. Balogun, T. Chakraborty, G. Muhawenayo, H. R. Kerner, A. M. Tárano, Z. Fang, N. Jacobs, S. Khanal, R. Abedi, and L. D. Estes, “Combining open labeled datasets with varying domains to improve large-scale agricultural field boundary delineation,” in *American Geophysical Union (AGU) Fall Meeting Abstracts*, Dec. 2025.

[6] G. Muhawenayo, Z. Fang, S. Khanal, A. Wollam, I. Corley, C. Robinson, M. Mohr, C. Holmes, J. Marcus, L. D. Estes, N. Jacobs, A. M. Tárano, and H. R. Kerner, “Global field boundary delineation model zoo and tooling,” in *American Geophysical Union (AGU) Fall Meeting Abstracts*, Dec. 2025.

[7] S. Shen, A. van Donkelaar, N. Jacobs, C. Li, and R. V. Martin, “Enhancing estimation of daily 1-km resolution fine particulate matter concentrations for north america with deep learning from geophysical a priori information,” in *American Geophysical Union (AGU) Fall Meeting Abstracts*, Dec. 2025. DOI: 10.1021/acsestair.5c00251

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[10] H. R. Kerner, S. Chaudhari, C. Robinson, A. Ghosh, A. Ahmad, E. Choi, N. Jacobs, C. Holmes, M. Mohr, and J. Marcus, "Fields of The World (FTW!): A new machine learning dataset for agricultural field boundary segmentation on four continents," in *American Geophysical Union (AGU) Fall Meeting Abstracts*, Dec. 2024.

[11] S. Sastry, A. Ahmad, A. Dhakal, S. Khanal, E. Xing, and N. Jacobs, "ClimSatDiff: Synthesizing the earth's surface conditioned on climatic variables using diffusion models," in *American Geophysical Union (AGU) Fall Meeting Abstracts*, Dec. 2024.

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[14] A. Dhakal, S. Khanal, S. Sastry, A. Ahmad, and N. Jacobs, "GeoBind: Binding text, image, and audio through satellite images," in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, (oral), Jul. 2024. arXiv: 2404.11720.

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[39] J. Zhu, A. M. Nolte, N. Jacobs, and M. Ye, "Incorporating machine learning with LiDAR for delineating sinkholes," in *Kentucky Water Resources Annual Symposium*, Mar. 2019.

[40] D. Jones, N. Jacobs, and S. Ellingson, "Learning deep feature representations for kinase polypharmacology," in *ACM Richard Tapia Celebration of Diversity in Computing Conference*, Sep. 2018.

[41] G. Liang, X. Wang, and N. Jacobs, “Evaluating the publicly available mammography datasets for deep learning model training,” in *SBI/ACR Breast Imaging Symposium*, Apr. 2018.

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[46] S. Workman, J. Knochelmann, N. Jacobs, D. S. White, and R. Hauer, “Registration and visualization of scientific aerial imagery at Kentucky Lake,” in *Kentucky EPSCoR Conference*, Oct. 2012.

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Technical Reports

[1] A. Abrams, C. Hawley, K. Miskell, A. Stoica, N. Jacobs, and R. Pless, “Shadow estimation method for "the episolar constraint: Monocular shape from shadow correspondence",” *arXiv*, vol. preprint 1304.4112 [cs.CV], Apr. 2013.

[2] N. Jacobs, S. Schuh, and R. Pless, “On unusual pixel shapes and image motion,” Computer Science and Engineering, Washington University in St. Louis, MO, USA, Tech. Rep. WUCSE-2009-16, Jun. 2009.

Datasets

[1] A. Abrams, J. Tucek, J. Little, N. Jacobs, and R. Pless, *LOST: Longterm observation of scenes (with tracks)*, http://mvrl.github.io/LOST.

[2] M. T. Islam, C. Greenwell, and N. Jacobs, *GeoFaces: A large database of geolocated face patches*, http://mvrl.github.io/GeoFaces.

[3] N. Jacobs, R. Pless, A. Abrams, and many others (see website for details), *AMOS: The archive of many outdoor scenes*, https://mvrl.github.io/AMOS.

[4] P. Mihail, S. Workman, Z. Bessinger, and N. Jacobs, *SkyFinder: A large dataset of webcam images annotated with sky regions*, https://mvrl.github.io/SkyFinder.

[5] M. U. Rafique, H. Blanton, and N. Jacobs, *Brooklyn Panorama Synthesis: A large dataset of panoramic images suitable for view synthesis evaluation.* https://mvrl.github.io/GAF.

[6] T. Salem, S. Workman, M. Zhai, and N. Jacobs, *Cross-View Time (CVT)*, https://mvrl.github.io/CVT.

[7] T. Salem, S. Workman, M. Zhai, and N. Jacobs, *Face2Year: A large number of images extracted from highschool yearbooks*, https://mvrl.github.io/Face2Year.

[8] S. Workman and N. Jacobs, *Cross-View ScenicOrNot (CVSoN)*, https://mvrl.github.io/CVSoN.

[9] S. Workman and N. Jacobs, *Crossview USA (CVUSA): A large dataset containing millions of pairs of ground-level and aerial/satellite images from across the United States.* https://mvrl.github.io/CVUSA.

[10] S. Workman, M. Zhai, and N. Jacobs, *Horizon Lines in the Wild (HLW): A large database of images with known horizon-line location*, http://mvrl.github.io/HLW.

Patents

[1] N. Jacobs and S. Workman, *Network architecture for generating a labeled overhead image*, US Patent App. 16/045,606, Jan. 2020.

[2] J. A. G. Whitney, J. T. Fessler, Z. C. N. Kratzer, N. B. Jacobs, A. M. Whitney, et al., *Method and system for estimating error in predicted distance using RSSI signature*, Jan. 2016.

5 Talks

1. "From Social Media to Sanderlings: Multimodal Tools for Understanding and Mapping Biodiversity" (Keynote), Dec 2025, Workshop on Imageomics @ NeurIPS, San Diego, CA

2. "Learning to Map Anything, Anywhere, Anytime" (Sanghi College of Engineering Dean's Special Invited Speaker), Dec 2025, Northern Arizona University, Flagstaff, AZ

3. "Learning to Map Anything, Anywhere, Anytime" (Invited Talk), Nov 2025, National Geospatial-Intelligence Agency, St. Louis, MO

4. "(Almost) Two Decades of Vision AcrosS alTitudes" (Keynote), Oct 2025, Workshop on 3D-VAST From street to space: 3D Vision AcrosS alTitudes (ICCV 2025 Workshop), Honolulu, Hawaii

5. "World Foundation Models for GeoAI" (Expert Panel), Sep 2025, Trillion Pixel Challenge Workshop, St. Louis, MO

6. "Toward Agentic GEOINT" (Lightning Talk), May 2025, GEOINT Symposium, St. Louis, MO

7. "Mapping and Multimodal Embeddings for Biodiversity" (Invited Talk), Mar 2025, Beyond Counting Trees (Bezos Earth Fund), Washington, DC

8. "Learning to Map Anything, Anywhere, Anytime" (Keynote), Nov 2023, The International Conference on Digital Image Computing: Techniques and Applications (DICTA), Port Macquarie, Australia

9. "Learning to Map Anything, Anywhere, Anytime" (Keynote), Oct 2023, ACM Multimedia Workshop (UAVs in Multimedia: Capturing the World from a New Perspective), Ottawa, Canada

10. "A Cost-Sensitive Approach To Dimensionality Reduction for Multispectral Imagery", Jul 2023, International Geoscience and Remote Sensing Symposium, Pasadena, CA

11. "Explorations in Self-Supervised Learning for Change Detection", Jul 2023, International Geoscience and Remote Sensing Symposium, Pasadena, CA

12. "Toward Dynamic Multimodal Remote Sensing: From Buildings and Populations to Soundscapes and Aesthetics", Apr 2023, Living Earth Collaborative, Washington University, St. Louis, MO

13. "Domain-Inspired Deep Learning for Computer Vision, Remote Sensing, and Medical Imaging", Oct 2022, Imaging Science Seminar, Washington University, St. Louis, MO

14. “Computer Vision for Multimodal Remote Sensing”, Aug 2022, WashU Geospatial Working Group Research Workshop, Washington University, St. Louis, MO
15. “A Structure-Aware Method for Direct Pose Estimation”, Jan 2022, IEEE Winter Conference on Applications of Computer Vision (WACV), Waikoloa Village, HI
16. Panelist for “Non-Traditional Careers in Computer Science” Nov 2021, ACM-W, University of Kentucky, Lexington, KY
17. “Mapping the Visual World Using Webcams, Cell Phones, and Satellites”, Oct 2021, Washington University in St. Louis, MO
18. “Learning Geo-Temporal Scene Models from Webcams, Cell Phones, and Satellites” (Keynote), Oct 2021, International Workshop on Distributed Smart Cameras, an ICCV Workshop (virtual)
19. “Mapping the Visual World Using Webcams, Cell Phones, and Satellites”, Dec 2020, University of Campinas, Unicamp, Brazil (virtual)
20. “Exploring the Intersection of Localization, Mapping, and Image Understanding” (Keynote), Aug 2020, ECCV Workshop on Long-Term Visual Localization (virtual)
21. “Deep Convolutional Neural Networks: Foundations to Frontiers (a 2-day short course)”, Mar 2020, Brazilian Space Agency (INPE), Sao Jose dos Campos, Brazil
22. “What, Where, and When: Mapping the World Using Webcams, Cell Phones, and Satellites”, Mar 2020, Brazilian Space Agency (INPE), Sao Jose dos Campos, Brazil
23. “Learning to Map Visual Appearance”, Feb 2020, Keeping Current Seminar, University of Kentucky (Computer Science), Lexington, KY
24. “Learning to Map Visual Appearance”, Jan 2020, Wageningen University, Netherlands
25. “What, Where, and When: Mapping the World Using Webcams, Cell Phones, and Satellites”, Nov 2019, University of Kentucky (Forestry), Lexington, KY
26. “Learning to Map the Visual World”, Jul 2019, Wright State University, Dayton, OH
27. “Understanding Places Using Ground-Level and Overhead Views” (Keynote), May 2019, Kentucky Geological Society (Annual Symposium), Lexington, KY
28. “Understanding Places Using Ground-Level and Overhead Views”, Feb 2019, Notre Dame University, South Bend, IN
29. “A Generative Model of Worldwide Facial Appearance” (Keynote), Jan 2019, Workshop on Demographic Variations in Performance of Biometric Algorithms, Waikoloa Village, HI
30. “A Generative Model of Worldwide Facial Appearance”, Jan 2019, IEEE Winter Conference on Applications of Computer Vision, Waikoloa Village, HI
31. “A Weakly Supervised Approach for Estimating Spatial Density Functions from High-Resolution Satellite Imagery”, Nov 2018, ACM SIGSPATIAL, Seattle, WA
32. “Understanding Places Using Ground-Level and Overhead Views”, Oct 2018, Commonwealth Computational Summit, Lexington, KY
33. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 4)”, Sep 2018, National Academy of Sciences (IC Academic Research Symposium), Washington, DC

34. “Understanding Places Using Ground-Level and Overhead Views”, Aug 2018, Oak Ridge National Lab, Oak Ridge, TN
35. “WhatGoesWhere: Predicting Object Distributions from Above”, Jul 2018, IGARSS, Valencia, Spain
36. “Building World Models for Situated Training and Planning”, May 2018, Air Force Science and Technology 2030 Workshop, Bloomington, IN
37. “Recent Advances in Image Understanding”, May 2018, DASC, Lexington, KY
38. “(Tutorial) Recent Advances in Deep Learning: Fusing Overhead and Ground-Level Views for Remote Sensing”, April 2018, USGIF Annual Symposium, Tampa, FL
39. “Understanding Places Using Ground-Level and Overhead Views”, Feb 2018, CVPR Area Chair Meeting, Toronto, Canada
40. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 3)”, Sep 2017, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
41. “GPU Accelerated Computer Vision, Remote Sensing, and Machine Learning”, Aug 2017, Kentucky Geological Service, Lexington, KY
42. “Fusing Overhead and Ground-Level Imagery to Improve Scene Understanding”, Jul 2017, Planet, San Francisco, CA
43. “Learning about When and Where from Imagery”, Jun 2017, Orbital Insight, Mountain View, CA
44. “(Tutorial) Recent Advances in Deep Learning: Fusing Overhead and Ground-Level Views for Remote Sensing”, Jun 2017, USGIF Annual Symposium, San Antonio, TX
45. “How Computers See People (extended)”, May 2017, CCTS Biomedical Informatics Seminar Series, Lexington, KY
46. “Understanding Places Using Ground-Level and Overhead Views”, May 2017, Midwest Vision Meeting, Chicago, IL
47. “How Computers See People”, Feb 2017, Suds’n’Science Speaker Series, West Sixth Brewing, Lexington, KY
48. “Learning about When and Where from Imagery”, Feb 2017, University of Missouri, Department of Computer Science
49. “Localization, Mapping, and Image Understanding”, Feb 2017, USGIF Machine Learning Symposium
50. “Deep Convolutional Neural Networks: Concepts and Examples (in Computer Vision”, Nov 2016, University of Kentucky, Society of Industrial and Applied Mathematics
51. “Crossview Convolutional Networks”, Oct 2016, Applied Imagery and Pattern Recognition, Washington, D.C.
52. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 2)”, Sep 2016, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
53. “Deep Convolutional Neural Networks: Concepts and Examples”, Jul 2016, University of Kentucky: Systems Biology and Omics Integration Seminar
54. “Crossview Methods for Localization and Mapping”, Jun 2016, IEEE CVPR Workshop on “Vision from Satellite to Street” (invited talk)
55. “A Fast Method for Estimating Transient Scene Properties”, Mar 2016, Winter Conference on Applications of Computer Vision, Lake Placid, NY

56. “Novel Cues for Geocalibration”, Feb 2016, Indiana University, Bloomington, IN
57. “Novel Cues for Camera Geocalibration”, Jan 2016, Uber Advanced Technology Center, Pittsburgh, PA
58. “Novel Cues for Geocalibration: Cloudy Days, Rainbows, and More”, Oct 2015, Carnegie Mellon University, Pittsburgh, PA
59. “Using Geotagged Internet Imagery to Understand the World”, Sep 2015, Université Laval, Quebec City, Canada
60. “face2gps: Estimating Geographic Location from Facial Features”, Sep 2015, International Conference on Image Processing, Quebec City, Canada
61. “GeoLookbook: Modeling Worldwide Human Visual Appearance”, Sep 2015, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
62. “Exploring the Geo-Dependence of Human Face Appearance”, Mar 2014, Winter Conference on Applications of Computer Vision, Steamboat Springs, CO
63. “Estimating Cloudmaps from Outdoor Image Sequences”, Mar 2014, Winter Conference on Applications of Computer Vision, Steamboat Springs, CO
64. “Scene Geometry from Several Partly Cloudy Days”, Oct 2013, International Conference on Distributed Smart Cameras, Palm Springs, CA
65. “Unlocking the Potential of the Global Network of Outdoor Webcams”, Apr 2013, Rochester Institute of Technology
66. “Geo-temporal Computer Vision: Applications to the NGA”, Nov 2011, National Geospatial-Intelligence Agency
67. “Geo-temporal Computer Vision: Applications to the Army”, Oct 2011, Army Research Lab
68. “Localizing, Calibrating, and Using Thousands of Outdoor Webcams”, Feb 2011, University of North Carolina–Charlotte
69. “Using Clouds Shadows to Infer Scene Structure and Camera Calibration”, Jun 2010, CVPR, San Francisco, CA
70. “Passive Vision and The Power of Collective Imaging”, Apr 2010, Object Video Inc., Reston, VA
71. “Localizing, Calibrating, and Using Thousands of Outdoor Webcams”, Apr 2010, University of Kentucky
72. “Time-Lapse Vision: Localizing, Calibrating, and Using Thousands Outdoor Webcams”, Apr 2010, Google, Mountain View, CA
73. “Passive Vision and The Power of Collective Imaging”, Jan 2010, Google, Mountain View, CA
74. “Incorporating Domain Constraints in Urban Vehicle Tracking”, Nov 2010, University of Missouri, Columbia, MO
75. “Compressive Sensing and Differential Image-Motion Estimation”, Mar 2010, ICASSP, Dallas, TX
76. “The Global Network of Outdoor Webcams: Properties and Applications ”, Nov 2009, ACM GIS, Seattle, WA
77. “Passive Vision: The Global Webcam Imaging Network”, Oct 2009, AIPR, Washington, DC
78. “Calibrating and Using the Global Network of Outdoor Webcams”, Aug 2009, ICDSC, Italy
79. “Adventures in Archiving and Using Three Years of Webcam Images”, Jun 2009, CVPR Workshop on Internet Vision, Miami, FL

80. “Recent Work: Webcams and Grooves”, Aug 2009, Object Video, Reston, VA
81. “Location-Specific Models for Tracking”, Jan 2008, WMVC, Copper Mountain, CO
82. “Using natural cues to geo-locate and geo-orient distributed cameras”, Jan 2008, VISN, Copper Mountain, CO
83. “Foreground Modeling: The Shape of Things That Came”, Feb 2007, WMVC, Austin, Texas

6 Service

University Service

- Washington University in St. Louis (2022–present)
  - 2022–present: Co-Director, Geospatial Research Initiative
  - 2022–present: Research Council, Taylor Geospatial Institute
  - 2023–present: Director of PhD Admissions, Department of Computer Science & Engineering
  - 2023–present: Faculty Technology Advisory Committee, James McKelvey School of Engineering
  - 2023–present: Curriculum Committee, Imaging Science Program
  - 2023–present: Faculty Advisor, WashU Robomaster Club
  - F2024: Faculty Search Committee, Imaging Science Program
  - F2023, F2024: Faculty Search Committee, Computer Science & Engineering Department
  - 2022–2023: Leadership Team, Geospatial Working Group
  - 2022–2023: Strategic Planning Steering Committee, James McKelvey School of Engineering
- University of Kentucky (2010–2022)
  - 2021–2022: Institute for Biomedical Informatics: Steering Committee
  - 2019–2022: Computer Science Department: Executive Committee
  - 2019–2022: College of Engineering: Master Planning/Space Committee
  - 2018–2019, 2020–2022: College of Engineering: Research Advisory Committee
  - 2020–2022: College of Engineering: Graduate Studies Team
  - 2013–2017, 2018–2022: Computer Science Department: Faculty Search Committee
  - 2020–2021: Computer Science Department: Chair Search Committee
  - 2020: College of Engineering: Recruiting Advisory Committee
  - 2018–2019: University Senate (Academic Facilities Committee, Technology Committee)
  - 2017: Member (Information Technology Task Force for Research Enablement and Outreach)
  - 2015–2016: Computer Science Department: ABET Committee
  - 2010–2012, 2015–2016: Computer Science Department: Media and Outreach
  - 2013: Center for Visualization and Virtual Environment: Director Search Committee
  - 2013: Computer Science Department: Chair Search Committee
  - 2012–2013: Computer Science Department: Curriculum Development Committee
  - 2012–2013, 2015: Engineering Day (oral presentation and/or software demonstration)

Professional Service

- Area Chair:
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2018, 2019, 2021, 2023, 2024 (senior), 2025 (lead), 2026–2027 (senior)]
  - European Conference on Computer Vision (ECCV) [2022, 2024 (lead)]
  - IEEE/CVF International Conference on Computer Vision (ICCV) [2023, 2025 (lead)]
  - Conference on Neural Information Processing Systems (NeurIPS) [2024]
  - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2014, 2022, 2023, 2027 (senior)]
- Organizing Committees:
  - IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2019–2023, 2025–2026]
  - Foundational Models Beyond the Visual Spectrum (workshop at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)) [2026]
  - Doctoral Consortium Co-Chair: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2017, 2019, 2024, 2025]
  - Industrial/Government Relations Chair: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2024]
  - Doctoral Consortium Chair: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2018, 2022]
  - Video Proceedings Chair: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2015]
  - IEEE Workshop on Motion and Video Computing (WMVC) [2011]
- Guest Editor:
  - ISPRS Journal of Photogrammetry and Remote Sensing [2024], Special Issue “Vision Language Models for Remote Sensing Analysis and Interpretation”
  - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS) [2021], Special Issue “Integrating User Generated Contents for Remote Sensing Applications”
  - Elsevier Computer Vision and Image Understanding (CVIU) [2019], Special Issue “Computer Vision for Remote Sensing”
- Panelist:
  - Roundtable Discussion at 1st Workshop on Computer Vision for Earth Observation (CV4EO) Applications 2024 (Hosted as part of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024)
- Session Chair:
  - IEEE International Geoscience and Remote Sensing Symposium (IGARSS) [2020,2023]
  - IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2019]
  - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2016, 2019, 2022, 2024]
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2018, 2024, 2025]

- IEEE/ACM International Conference on Distributed Smart Cameras [2013]

- Reviewing for Journals:
  - Proceedings of the National Academy of Sciences [2024]
  - IEEE Transactions on Geoscience and Remote Sensing [2017, 2020]
  - ISPRS Journal of Photogrammetry and Remote Sensing [2019, 2020]
  - IEEE Transactions on Pattern Analysis and Machine Intelligence [2011×2, 2012, 2018, 2019]
  - IEEE Transactions on Multimedia [2011, 2016]
  - Elsevier Computer Vision and Image Understanding [2010, 2013, 2016×2]
  - IEEE Transactions on Computational Imaging [2016]
  - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS) [2015]
  - Springer Machine Vision and Applications [2014]
  - IEEE Sensors [2014]
  - Elsevier Image and Vision Computing [2013]
  - IEEE Transactions on Circuits and Systems for Video Technology [2007–2011]
  - IEEE Computer Graphics and Applications [2010]
  - IEEE Transactions on Aerospace and Electronic Systems [2010]
  - Elsevier Computers and Electronics in Agriculture [2010]
  - Cartography and Geographic Information Science [2010]

- Program Committee / Reviewer for:
  - Conferences
    * IACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR) [2026]
    * International Conference on Learning Representations (ICLR) [2022, 2025]
    * IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2021, 2024, 2025]
    * IEEE International Geoscience and Remote Sensing Symposium (IGARSS) [2020]
    * British Machine Vision Conference (BMVC) [2020]
    * IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2006–2017, 2020]
    * European Conference on Computer Vision (ECCV) [2010, 2014, 2020]
    * Neural Information Processing Systems (NeurIPS) [2010–2012, 2020]
    * AAAI Conference on Artificial Intelligence (AAAI) [2020, 2026 (social impact track)]
    * IEEE/CVF International Conference on Computer Vision (ICCV) [2007, 2009, 2019, 2021]
    * Asian Conference on Computer Vision (ACCV) [2010, 2016]
    * IEEE International Conference on Robotics and Automation (ICRA) [2016]
    * International Conference on Machine Learning (ICML) [2012]
    * IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS) [2010]
  - Workshops
    * ECCV Workshop TerraBytes — Towards global datasets and models for Earth observation [2026]
    * ICLR Workshop on Machine Learning for Remote Sensing [2023]
    * IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2017, 2019, 2020]
    * CVPR Workshop on Photogrammetric Computer Vision [2019]

    * CVPR Workshop on DeepGlobe Satellite Challenge [2018]
    * ACM International Workshop on Geotagging and Its Applications [2013]
    * ICCV Workshop on Computer Vision for Converging Perspectives [2013]
    * IEEE Workshop on Applications of Computer Vision [2012–2013]
    * ECCV Workshop on Visual Analysis and Geo-Localization of Large-Scale Imagery [2012]
    * ACM Workshop on Geotagging and Its Applications in Multimedia [2012]
    * IEEE Workshop on Motion and Video Computation [2009–2011]

- Reviewing for Funding Agencies:
  - Panelist for NSF Information and Intelligent Systems Division [2015, 2016, 2017, 2019]
  - Panelist for NSF Division of Industrial Innovation and Partnerships [2016]
  - External reviewer for NSF Information and Intelligent Systems Division [2015]
  - External reviewer for Fonds de recherche du Quebec [2014]
- Advisory Committee for:
  - 1st Workshop on Computer Vision for Earth Observation (CV4EO) Applications 2024 (Hosted as part of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024)

Memberships

- Senior Member: Institute of Electrical and Electronics Engineers
- Full Member: British Machine Vision Association and Society for Pattern Recognition
- Affiliate Member: International Association of Pattern Recognition

Graduate Committee Service

Ph.D. Committees

1. David Sundarsingh (Electrical and Systems Engineering, TBD)
2. Timur Kasimov (Computer Science, TBD)
3. Yuanlong Su (EECE, TBD)
4. Chengsong Huang (Computer Science, TBD)
5. Ben Warner (Computer Science, TBD)
6. Evelyn Yang (Computer Science, TBD)
7. Sidrah Liaqat (Electrical Engineering, TBD)
8. Yu Yan (EECE, TBD)
9. Lisa Liao (Computer Science, TBD)
10. Anindya Sarkar (Computer Science, TBD)
11. Nischal Khanal (Imaging Science, TBD)
12. Lydia Reader (2026, Computational & Data Sciences)
13. Siyuan Shen (Energy, Environmental & Chemical Engineering, 2026)
14. Ruiqi Wang (Computer Science, 2026)
15. Tri Pham (Computer Science, 2025)
16. Sheng-Chieh Lin (Physics, 2025)
17. Weijie Gan (Computer Science, 2025)
18. Yiwen Ju (Computer Science, 2025)
19. Hao Liu (Computer Science, 2025)
20. Junlin Wu (Computer Science, 2025)
21. Gustavo Gratacós (Computer Science, 2024)
22. Ye Htet (Computer Science, 2024)

23. Minoo Hosseinzadeh (Computer Science, 2024)
24. Stephen Parsons (Computer Science, 2023)
25. Md Selim (Computer Science, 2023)
26. Ashutosh Timilsina (Computer Science, 2023)
27. Arnab Sarkar (Physics, 2022)
28. David Adeniji (Mechanical Engineering, 2022)
29. Tarannum Shaila Zaman (Computer Science, 2022)
30. Chengxi Li (Computer Science, 2022)
31. Sajad Javadinasab Hormozabad (Civil Engineering, 2021)
32. Alireza Shirvani (Computer Science, 2021)
33. Narjes Bozorg (Electrical Engineering, 2020)
34. Kyle Helfrich (Math, 2020)
35. Sifei Han (Computer Science, 2019)
36. Xinxin Zuo (Computer Science, 2019)
37. Jinping Zhuge (Math, 2019)
38. Nkiruka Uzuegbunam (Electrical Engineering, 2018)
39. Anthony Rios (Computer Science, 2018)
40. Po-Chang Su (Electrical Engineering, 2017)
41. Yajie Zhao (Computer Science, 2017)
42. Wesley Hough (Computer Science, 2017)
43. Hasan Sajid (Electrical Engineering, 2016)
44. Harikrishnan Unnikrishnan (Electrical Engineering, 2015)
45. Bo Fu (Computer Science, 2015)
46. Shaoceng Wei (Statistics, 2015)
47. Yan Huang (Computer Science, 2014)
48. Mao Ye (Computer Science, 2014)
49. Chenxi Zhang (Computer Science, 2014)
50. Ju Shen (Electrical Engineering, 2014)

External Ph.D. Committees

1. Rachel Badzioch (Notre Dame University, Biological Sciences, TBD)
2. Christoph Gerhardt (Technische Universität Ilmenau, Computer Science, TBD)
3. Morris Alper (Tel Aviv University, Computer Science, 2025)
4. Eike Jens Hoffmann (Technical University of Munich, Data Science in Earth Observation, 2022)
5. Ahmed Nassar (IRISA, Université Bretagne Sud, Vannes, Computer Science, 2021)
6. Céline Portenier (University of Bern, Computer Science, 2021)
7. Raian Maretto (National Institute for Space Research, Geoinformation Science, 2020)
8. Shivangi Srivastava (Wageningen University, Netherlands, Computer Science, 2020)
9. Nam Vo (Georgia Institute of Technology, Computer Science, 2019)
10. Yannick Hold-Geoffroy (Laval University, Quebec, CA, Computer Science, 2018)
11. Ethan Welty (University of Colorado–Boulder, Environmental Studies, 2018)

Master's Committees

1. Yiding Yang (Computer Science, 2026)
2. Aadarsha Gopala Reddy (Computer Science, 2026)
3. Aaron Luo (Computer Science, 2026)
4. Stuart Aldrich (Computer Science, 2026)
5. Weining Wang (Computer Science, 2026)
6. Kyle Wolford (Computer Science, 2025)
7. Haris Naveed (Computer Science, chair, 2025)
8. Yuxuan Yang (Computer Science, 2025)

9. Zifan Wang (Computer Science, 2025)
10. Sizhe Zhang (Statistics, 2024)
11. Yin Li (Computer Science, 2024)
12. Evin Jaff (Computer Science, 2024)
13. Owen Ma (Computer Science, 2024)
14. Patrick Lynch (Computer Science, 2024)
15. Joshua Tang (Computer Science, 2024)
16. Kyle Montgomery (Computer Science, 2024)
17. Fiona Xu (Computer Science, 2023)
18. Emma McMillian (Computer Science, 2023)
19. Chang Ti (Computer Science, 2023)
20. Brian Chao (Computer Science, 2023)
21. Yihang Xu (Computer Science, 2023)
22. David Sarpong (Computer Science, 2023)
23. Peizhen Tong (Computer Science, 2023)
24. Di Huang (Computer Science, 2023)
25. Nan Huang (Computer Science, 2023)
26. Zihao Zou (Computer Science, 2022)
27. Zhou Chu (Computer Science, 2022)
28. Aiden McIlraith (Computer Science, 2022)
29. Yuan Liu (Computer Science, 2022)
30. Subash Khanal (Electrical Engineering, 2020)
31. Genghis Goodman (Computer Science, 2019)
32. Jonathan Dingess (Computer Science, 2019)
33. Ryan Zembrodt (Computer Science, 2019)
34. Xiaofei Zhang (Computer Science, 2017)
35. Qingguo Xu (Computer Science, 2017)
36. DhiShankar Bhattacharya (Computer Science, 2017)
37. Stanley Rosenbaum (Computer Science, 2016)
38. Sean Karlage (Computer Science, 2016)
39. Hasan Sajid (Electrical Engineering, 2014)
40. Edwin Prem Kumar Sathiyamoorthy (Electrical Engineering, 2011)

7 Teaching

Courses Taught

The following list summarizes the traditional, classroom courses I have taught:

- *Advances in Computer Vision*, CSE 659a/5519, [F2024,F2025], Washington University in St. Louis
- *Computer Vision*, CSE 559a/5509, [S2023, S2024, S2025, S2026], Washington University in St. Louis
- *Machine Learning*, CS 460g, [F2012, F2013, F2014, F2016, F2018, F2019], University of Kentucky
- *Computer Vision*, CS 636, [S2011, S2013, S2017], University of Kentucky
- *Learning-Based Methods for Computer Vision*, CS 585/685, [S2015], University of Kentucky
- *Advanced Topics in Computer Science: Machine Learning*, CS 685, [S2012], University of Kentucky
- *Intermediate Topics in Computer Science: Computational Photography*, CS 585, [F2010, F2011], University of Kentucky
- *Theory of Computation*, CECS 341, [F2002], University of Missouri

8 Lab Members / Mentees

Postdoctoral Scholars

1. Subash Khanal — Dates: 2025–present
   Research Focus: Geospatial Representation Learning, Remote Sensing
2. Adeel Ahmad (Ph.D. Geomatics, University of Punjab–Lahore) — Dates: 2023–2025
   Research Focus: Remote Sensing, Deep Learning, Land-Use Modeling
3. Benjamin Brodie (Ph.D. Mathematics, University of Kentucky) — Dates: 2020–2022
   Research Focus: Change Detection, Object Tracking, Re-Identification, Metric Learning

Research Staff

1. Hamza Iqbal (MS, Computer Science, Washington University in St. Louis) — Dates: 2026–present
   Role: Research Associate
2. Jonathan Lin (MS, Computer Science, Washington University in St. Louis) — Dates: 2024–present
   Role: Research Associate

Ph.D. Students

1. Chih-Chuan (Leo) Hsu — Degree: Ph.D., Computer Science
   Title: TBD — Date: TBD
2. Brian Wei — Degree: Ph.D., Computer Science
   Title: TBD — Date: TBD
3. Danni Beaulieu — Degree: Ph.D., Computer Science
   Title: TBD — Date: TBD
4. Tianyi Gao — Degree: Ph.D., Computer Science
   Title: TBD — Date: TBD
5. Feng Qiao — Degree: Ph.D., Computer Science
   Title: TBD — Date: TBD
6. Carlos Lopez de la Cerda Bazaldu [*co-chair* with Jacob Montgomery] — Degree: Ph.D., Computational & Data Sciences
   Title: TBD — Date: TBD
7. Daniel Cher — Degree: Ph.D., Computational & Data Sciences
   Title: TBD — Date: TBD
8. Eric Xing — Degree: Ph.D., Computer Science
   Title: TBD — Date: TBD
9. Tong Li [*co-chair* with Joshua Oltmanns] — Degree: Ph.D., Computational & Data Sciences
   Title: TBD — Date: TBD
10. Nia Hodges — Degree: Ph.D., Systems Science and Mathematics
    Title: TBD — Date: TBD
11. Alex Wollam — Degree: Ph.D., Computer Science
    Title: TBD — Date: TBD

12. Zhexiao Xiong — Degree: Ph.D., Computer Science
    Title: TBD — Date: May 2027 (est)

13. Michael Lanier [*co-chair* with Yevgeniy Vorobeychik] — Degree: Ph.D., Computer Science
    Title: TBD — Date: May 2027 (est)

14. Srikumar Sastry — Degree: Ph.D., Imaging Science
    Title: Task-Aligned Multimodal Representation Learning — Date: Dec 2026 (est)

15. Aayush Dhakal — Degree: Ph.D., Computer Science
    Title: TBD — Date: Dec 2026 (est)

16. Subash Khanal — Degree: Ph.D., Computer Science
    Title: Multimodal Representation Learning for Geospatial Soundscape Mapping — Date: Jul 2025
    Employment: Postdoctoral Scholar, Washington University in St. Louis

17. Xin Xing [*co-chair* with Ai-Ling Lin] — Degree: Ph.D., Computer Science
    Title: Structured Attention for Image Analysis — Date: Nov 2023
    Employment: Assistant Professor, University of Nebraska-Omaha

18. Yu Zhang — Degree: Ph.D., Computer Science
    Title: Multimodal Domain Generalization — Date: Mar 2023
    Employment: Assistant Professor, Computer Science, Boise State University

19. Connor Greenwell — Degree: Ph.D., Computer Science
    Title: Probabilistic Cross-Domain Representation Learning — Date: Jun 2022
    Employment: Senior R&D Engineer, Kitware Inc.

20. Hunter Blanton — Degree: Ph.D., Computer Science
    Title: Revisiting Absolute Pose Regression — Date: Aug 2021
    Employment: Senior Computer Vision Engineer, Yembo (startup)

21. Usman Rafique [*co-chair* with Samson Cheung] — Degree: Ph.D., Electrical Engineering
    Title: Weakly Supervised Learning for Multi-Image Synthesis — Date: Jul 2021
    Employment: Research Scientist, Kitware Inc.

22. Gongbo "Tony" Liang — Degree: Ph.D., Computer Science
    Title: Clinical-Inspired Multi-Modal Deep Learning Medical Imaging Analysis — Date: Oct 2020
    Employment: Assistant Professor, Computer Science, Eastern Kentucky University

23. Tawfiq Salem — Degree: Ph.D., Computer Science
    Title: Learning to Map the Visual and Auditory World — Date: Jul 2019
    Employment: Visiting Assistant Professor, Computer and Information Technology, Purdue University

24. Zach Bessinger — Degree: Ph.D., Computer Science
    Title: Modeling and Mapping Location-Dependent Human Appearance — Date: Dec 2018
    Employment: Senior Applied Scientist, Zillow

25. Menghua "Ted" Zhai — Degree: Ph.D., Computer Science
    Title: Deep Probabilistic Models for Camera Geo-Calibration — Date: Dec 2018
    Employment: Computer Vision Engineer, MatrixTime (startup)

26. Scott Workman — Degree: Ph.D., Computer Science
    Title: Leveraging Overhead Imagery for Localization, Mapping, and Understanding — Date: Apr 2018
    Employment: Research Scientist, DZYNE Technologies

27. Hamid Hamraz — Degree: Ph.D., Computer Science
Title: Computational Forest Modeling using Airborne Remote Sensing LiDAR — Date: Apr 2018
Employment: Computational and Data Scientist, Microsoft

28. Mohammad T. Islam — Degree: Ph.D., Computer Science
Title: Analyzing the Geo-Dependence of Human Face Appearance and Its Applications — Date: Jul 2016
Employment: Associate Professor, Computer Science, Southern Connecticut State University

29. Paul Mihail [*co-chair* with Judy Goldsmith] — Degree: Ph.D., Computer Science
Title: Visualizing and Predicting the Effects of Rheumatoid Arthritis on Hands — Date: May 2014
Employment: Professor, Computer Science, Valdosta State University

Visiting Scholars

1. Abby Stylianou (Saint Louis University) [*Visiting Associate Professor*] — Dates: 2025–2026
Topic: TBD

2. Clemens Mosig (Leipzig University) — Dates: 2025
Topic: Deep Learning for Tree Mortality Detection

3. José Nascimento (UNICAMP, Brazil) — Dates: 2024
Topic: TBD

4. Alex Levering (Wageningen University) — Dates: 2022–2023
Topic: Landscape Quality Assessment

5. Rafael Padilha (UNICAMP, Brazil) — Dates: 2019–2020
Topic: Image Forensics

6. Raian Vargas Maretto (INPE, Brazil) — Dates: 2018–2019
Topic: Deforestation Detection

7. Patrick Tutzauer (University of Stuttgart) — Dates: Fall 2017
Topic: Geospatial Trajectory Modeling

Master's Students

1. Rukai Gao — Degree: MS, Computer Science
Title: 3D Laryngeal Tumor Mapping via Near-Field SLAM — Date: Fall 2026

2. Mohammad Rouie Miab — Degree: MS, Computer Science
Title: Improving Semantic Precision in Text-to-Image Diffusion Models via Latent-Space Optimization and Semantically-Parsed Evaluation — Date: Spring 2026

3. Haris Naveed — Degree: MS, Computer Science
Title: Deep Learning for Tree Canopy Height Estimation — Date: Spring 2025

4. Jackson McCall — Degree: MS, Computer Science
Title: GeoSynth++: Large-Scale Satellite Image Synthesis — Date: Spring 2025

5. Jingyun Ma — Degree: MS, Computer Science
Title: Large Language Models for MCNP Input Generation — Date: Spring 2025

6. Shuhan (Steven) Zhang — Degree: MS, Computer Science
Title: Query-Specific Feature Transformation for Fine-Grained Image Retrieval — Date: Spring 2025

7. Vinh Pham — Degree: MS, Engineering Data Analytics & Statistics
   Title: Automated Satellite Imagery Analysis for Global Agricultural Field Boundary Detection — Date: Spring 2025

8. Ethan Weilheimer — Degree: MS (thesis), Computer Science
   Title: Partially Supervised Reinforcement Learning for GPS-Denied Navigation — Date: Spring 2025

9. Myan Sudharsanan — Degree: MS, Computer Engineering
   Title: VisionLLMs in the Automotive Domain: Finegrained Details of Traffic Signs and Passenger Vehicles — Date: Spring 2025

10. Lunchi Guo — Degree: MS, Computer Engineering
    Title: MedVid-Align: Intelligent Analysis of Medical Procedure Videos — Date: Spring 2025

11. Lunchi Guo — Degree: MS, Computer Engineering
    Title: Deep Learning-Based Coordinate Prediction from Medical Fluoroscopic Images for Improved Radiation — Date: Fall 2024

12. Wanzhou Liu — Degree: MS, Computer Science
    Title: High Efficiency Generalizable Driving World Model — Date: Fall 2024

13. Nia Hodges — Degree: MS, Engineering Data Analytics & Statistics
    Title: Wide-Area Image Localization — Date: May 2024

14. Hongzhang Wang — Degree: MS, Computer Science
    Title: Monocular Depth Estimation — Date: Dec 2023

15. Alex Wollam — Degree: MS, Computer Science
    Title: Exploring Sequential Outdoor Panorama Synthesis with Diffusion Models — Date: Jul 2023

16. Alex Greene — Degree: MS, Computer Science
    Title: Using Aerial Imagery to Estimate Ground-level Object Distributions — Date: May 2023

17. Zachary Pulliam — Degree: MS, Data Science
    Title: TBD — Date: 2022

18. Jacob Birge — Degree: MS, Computer Science
    Title: A Cost-Sensitive Approach To Multimodal Fusion — Date: Dec 2021

19. David T. Jones — Degree: MS, Computer Science
    Title: Intensity Harmonization for Airborne LiDAR — Date: May 2021

20. Armin Hadzic — Degree: MS, Computer Science
    Title: Estimating Free-Flow Speed with LiDAR and Overhead Imagery — Date: May 2020

21. Weilian "William" Song — Degree: MS, Computer Science
    Title: Image-Based Roadway Assessment using Convolutional Neural Networks — Date: May 2019

22. William "Derek" Jones [*co-chair* with Sally Ellingson] — Degree: MS, Computer Science
    Title: Scalable Feature Selection and Extraction with Applications in Kinase Polypharmacology — Date: May 2018

23. Ryan Baltenberger — Degree: MS, Computer Science
    Title: Estimating Transient Scene Attributes Using Deep Convolutional Neural Networks — Date: May 2016

24. Feiyu Shi — Degree: MS, Electrical Engineering
    Title: Principal Component Analysis For Multi-size Images — Date: Dec 2013

Undergraduate Research Students

1. Kexing (Ariana) Li — Dates: Fall 2025
   Title: TBD
2. Phoenix Jarosz — Dates: Fall 2025
   Title: TBD
3. Rowan Sandvig — Dates: Fall 2025
   Title: TBD
4. Nick Haisler — Dates: 2025 (REU, Drake University)
   Title: TBD
5. Ice Cui — Dates: Fall 2025
   Title: Leveraging Diffusion Transformers to Improve GeoSynth Scalability and Cross-Scale Consistency
6. Dev Gupta — Dates: Fall 2025
   Title: Temporally-Consistent Stereo Video Generation
7. Brian Wei — Dates: Fall 2025
   Title: Stereo Video Analysis
8. Brian Song — Dates: 2024 (REU, Vanderbilt University)
   Title: TBD
9. Gabriella Ramirez — Dates: 2024 (REU, University of Missouri)
   Title: TBD
10. John Lim — Dates: 2024 (REU, Northwestern University)
    Title: TBD
11. Eddie Choi — Dates: 2024 (Post-Bacc Research Assistant)
    Title: TBD
12. Daniel Khen — Dates: 2024
    Title: TBD
13. Sam Schwartz — Dates: 2023
    Title: TBD
14. Josh Liu — Dates: 2023
    Title: TBD
15. Cohen Archbold — Dates: 2020–2022
    Title: Automatic Real-Estate Price Estimation
16. Matthew Mitchell — Dates: 2022
    Title: Remote Sensing for Social Good
17. Gareth Walker — Dates: 2022
    Title: Remote Sensing for Social Good
18. Julia Stekardis — Dates: 2021–2022
    Title: Large-Scale Image Geo-Localization
19. Evan Bolton — Dates: 2021
    Title: Generating Synthetic Training Data using a Game Engine

20. Shashank Bhatt — Dates: 2019–2022
    Title: Multi-Object Tracking
21. Nicole Wong — Dates: 2021
    Title: TBD
22. Sean Grate — Dates: 2019–2020
    Title: Deep Learning for Point Clouds
23. Evan Yang — Dates: 2019
    Title: TBD
24. Thomas Barber — Dates: 2019
    Title: Deep Learning for Remote Sensing
25. Yuhan Long — Dates: 2019
    Title: Deep Learning for Medical Imaging
26. Weilian Song — Dates: 2016–2019
    Title: Applications of Deep Convolutional Neural Networks to Geometric Computer Vision
27. Aaron Mueller — Dates: 2018
    Title: Deep Learning for Educational Data
28. Sam Davidson — Dates: 2016–2017
    Title: Applications of Generative Adversarial Networks to Social Media Imagery
29. Connor Greenwell — Dates: 2014–2016
    Title: Interactive Methods for Aerial Imagery Understanding
30. Ryan Baltenberger — Dates: 2012–2015
    Title: Understanding Outdoor Scene Appearance
31. Angelo Stekardis — Dates: 2014–2015
    Title: Understanding Facial Expressions
32. J. David Smith — Dates: 2013–2015
    Title: User-in-the-loop Camera Calibration
33. Noora Aljabi — Dates: 2013
    Title: Using Flickr to Map Phenological Trends
34. Kyle Kolpek — Dates: 2012
    Title: Aerial Image Registration
35. Jim Knochelmann — Dates: 2011–2012
    Title: User-Tools for Aerial Image Registration

High School Research Students

1. Krishna Bhatraju — Dates: 2021–2022
   Title: Deep Motion Estimation
2. William Greenlee — Dates: 2021–2022
   Title: Deep Learning for Computer Vision
3. Chris Wang — Dates: 2019–2021
   Title: Multimodal Medical Imaging for Alzheimer's Disease Classification

4. Nicole Wong — Dates: 2019–2020
   Title: Learning-Based View Synthesis
5. Whitney Dawson — Dates: 2019
   Title: TBD
6. Cohen Archbold — Dates: 2017–2018
   Title: Photo-Geolocation using Convolutional Neural Networks
7. Ryan Landry — Dates: 2017–2018
   Title: RRADCL: Rapid Roadway Assessment with Deep Convolutional Learning
8. C. J. Labianca — Dates: 2016–2017
   Title: Evaluation of Optimization Algorithms for Deep Convolutional Neural Networks
9. Andrew Albrecht — Dates: 2016–2017
   Title: Mapping Social Media Imagery
10. Andrew Tapia — Dates: 2014–2015
    Title: Estimating Surface Reflectivity
11. Alex Lucas — Dates: 2013–2014
    Title: Evaluation of Automatic Face Detection Methods
12. Ryan Baltenberger — Dates: 2011–2012
    Title: Gesture-Based User Interaction with the Microsoft Kinect